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TD N°4 – ADMINISTRATION DES SYSTÈMES INFORMATIQUES I

Niveau : L3

Date : Le/3/18/26

L’objectif de ce TD est de :

1. analyser les ports ouverts
2. configurer un firewall simple
3. écrire des règles de filtrage réseau
4. comprendre les chaînes de filtrage
5. comparer différentes solutions de firewall Linux
6. analyser une configuration SSH
7. sécuriser les accès
8. mettre en place l’authentification par clé
9. limiter les connexions SSH
10. appliquer les bonnes pratiques de sécurité

Travaux Dirigés

1. Reproduire les tâches 1 à 12

1)Analyse du serveur

```
Windows PowerShell    doche@ubuntuserve: ~

doche@ubuntuserve:~$ ss -tuln
Netid  State   Recv-Q   Send-Q                      Local Address:Port
                   Peer Address:Port                Process
udp    UNCONN  0        0                                127.0.0.1:161
                       0.0.0.0:*
udp    UNCONN  0        0                               127.0.0.54:53
                       0.0.0.0:*
udp    UNCONN  0        0                            127.0.0.53%lo:53
                       0.0.0.0:*
udp    UNCONN  0        0                        172.20.10.2%enp0s3:68
                       0.0.0.0:*
udp    UNCONN  0        0                                  0.0.0.0:111
                       0.0.0.0:*
udp    UNCONN  0        0                                    [::1]:161
                         [::]:*
udp    UNCONN  0        0          [fe80::a00:27ff:fe62:e94a]%enp0s3:546
                         [::]:*
udp    UNCONN  0        0                                     [::]:111
                         [::]:*
tcp    LISTEN  0        4096                               0.0.0.0:22
                       0.0.0.0:*
tcp    LISTEN  0        100                                0.0.0.0:25
                       0.0.0.0:*
tcp    LISTEN  0        4096                               0.0.0.0:111
                       0.0.0.0:*
tcp    LISTEN  0        511                                0.0.0.0:80
                       0.0.0.0:*
tcp    LISTEN  0        151                              127.0.0.1:3306
                       0.0.0.0:*
tcp    LISTEN  0        4096                            127.0.0.54:53
                       0.0.0.0:*
tcp    LISTEN  0        4096                         127.0.0.53%lo:53
                       0.0.0.0:*
tcp    LISTEN  0        70                               127.0.0.1:33060
                       0.0.0.0:*
tcp    LISTEN  0        4096                               0.0.0.0:10050
                       0.0.0.0:*
tcp    LISTEN  0        4096                                  [::]:22
                         [::]:*
tcp    LISTEN  0        100                                   [::]:25
                         [::]:*
tcp    LISTEN  0        4096                                  [::]:111
                         [::]:*
tcp    LISTEN  0        511                                   [::]:80
                         [::]:*
tcp    LISTEN  0        4096                                  [::]:10050
```

```
doche@ubuntuserve:~$ systemctl list-units --type=service
  UNIT                                      LOAD   ACTIVE SUB     DESCRIPTIO>
  apache2.service                           loaded active running The Apache>
  apparmor.service                          loaded active exited  Load AppAr>
  apport.service                            loaded active exited  automatic >
  blk-availability.service                  loaded active exited  Availabili>
  console-setup.service                     loaded active exited  Set consol>
  cron.service                              loaded active running Regular ba>
  dbus.service                              loaded active running D-Bus Syst>
  finalrd.service                           loaded active exited  Create fin>
  getty@tty1.service                        loaded active running Getty on t>
  keyboard-setup.service                    loaded active exited  Set the co>
  kmod-static-nodes.service                 loaded active exited  Create Lis>
  lvm2-monitor.service                      loaded active exited  Monitoring>
  ModemManager.service                      loaded active running Modem Mana>
  multipathd.service                        loaded active running Device-Map>
  mysql.service                             loaded active running MySQL Comm>
  networkd-dispatcher.service               loaded active running Dispatcher>
  nginx.service                             loaded active running A high per>
  plymouth-quit-wait.service                loaded active exited  Hold until>
  plymouth-quit.service                     loaded active exited  Terminate >
  plymouth-read-write.service               loaded active exited  Tell Plymo>
  polkit.service                            loaded active running Authorizat>
  postfix.service                           loaded active exited  Postfix Ma>
  postfix@-.service                         loaded active running Postfix Ma>
  rpcbind.service                           loaded active running RPC bind p>
  rsyslog.service                           loaded active running System Log>
  setvtrgb.service                          loaded active exited  Set consol>
  snapd.apparmor.service                    loaded active exited  Load AppAr>
  snapd.seeded.service                      loaded active exited  Wait until>
  snmpd.service                             loaded active running Simple Net>
  ssh.service                               loaded active running OpenBSD Se>
  sysstat.service                           loaded active exited  Resets Sys>
  systemd-binfmt.service                    loaded active exited  Set Up Add>
  systemd-journal-flush.service             loaded active exited  Flush Jour>
  systemd-journald.service                  loaded active running Journal Se>
  systemd-logind.service                    loaded active running User Login>
  systemd-modules-load.service              loaded active exited  Load Kerne>
  systemd-networkd-wait-online.service      loaded active exited  Wait for N>
  systemd-networkd.service                  loaded active running Network Co>
  systemd-random-seed.service               loaded active exited  Load/Save >
  systemd-remount-fs.service                loaded active exited  Remount Ro>
  systemd-resolved.service                  loaded active running Network Na>
  systemd-sysctl.service                    loaded active exited  Apply Kern>
  systemd-timesyncd.service                 loaded active running Network Ti>
```

2. Firewall simple avec UFW

```
doche@ubuntuserve:~$ sudo apt install ufw -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
ufw est déjà la version la plus récente (0.36.2-6).
ufw passé en « installé manuellement ».
0 mis à jour, 0 nouvellement installés, 0 à enlever et 9 non mis à jour.
```

```
doche@ubuntuserve:~$ sudo ufw allow 22
Rule added
Rule added (v6)
doche@ubuntuserve:~$ sudo ufw allow 80
Rule added
Rule added (v6)
doche@ubuntuserve:~$ sudo ufw allow 443
Rule added
Rule added (v6)
doche@ubuntuserve:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
22                         ALLOW IN    Anywhere
80                         ALLOW IN    Anywhere
443                        ALLOW IN    Anywhere
22/tcp (v6)                ALLOW IN    Anywhere (v6)
22 (v6)                    ALLOW IN    Anywhere (v6)
80 (v6)                    ALLOW IN    Anywhere (v6)
443 (v6)                   ALLOW IN    Anywhere (v6)

doche@ubuntuserve:~$
```

3. Configuration avec iptables

doche@ubuntuserve: ~

```
443 (v6)                   ALLOW IN    Anywhere (v6)

doche@ubuntuserve:~$ sudo iptables -L
Chain INPUT (policy DROP)
target     prot opt source               destination
ufw-before-logging-input  all  --  anywhere             anywhere
ufw-before-input  all  --  anywhere             anywhere
ufw-after-input  all  --  anywhere             anywhere
ufw-after-logging-input  all  --  anywhere             anywhere
ufw-reject-input  all  --  anywhere             anywhere
ufw-track-input  all  --  anywhere             anywhere

Chain FORWARD (policy DROP)
target     prot opt source               destination
ufw-before-logging-forward  all  --  anywhere             anywhere

ufw-before-forward  all  --  anywhere             anywhere
ufw-after-forward  all  --  anywhere             anywhere
ufw-after-logging-forward  all  --  anywhere             anywhere

ufw-reject-forward  all  --  anywhere             anywhere
ufw-track-forward  all  --  anywhere             anywhere

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
ufw-before-logging-output  all  --  anywhere             anywhere

ufw-before-output  all  --  anywhere             anywhere
ufw-after-output  all  --  anywhere             anywhere
ufw-after-logging-output  all  --  anywhere             anywhere
ufw-reject-output  all  --  anywhere             anywhere
ufw-track-output  all  --  anywhere             anywhere

Chain ufw-after-forward (1 references)
target     prot opt source               destination

Chain ufw-after-input (1 references)
target     prot opt source               destination
ufw-skip-to-policy-input  udp  --  anywhere             anywhere
 udp dpt:netbios-ns
ufw-skip-to-policy-input  udp  --  anywhere             anywhere
 udp dpt:netbios-dgm
ufw-skip-to-policy-input  tcp  --  anywhere             anywhere
 tcp dpt:netbios-ssn
ufw-skip-to-policy-input  tcp  --  anywhere             anywhere
 tcp dpt:microsoft-ds
```

```
doche@ubuntuserve:~$ sudo iptables -P INPUT DROP
doche@ubuntuserve:~$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
doche@ubuntuserve:~$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
doche@ubuntuserve:~$ sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
doche@ubuntuserve:~$ sudo iptables -A INPUT -m conntrack --ctstate ESTABLISH
ED,RELATED -j ACCEPT
doche@ubuntuserve:~$
```

4. Firewall moderne avec nftables

```
doche@ubuntuserve:~$ sudo apt install nftables -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
nftables est déjà la version la plus récente (1.0.9-1ubuntu0.1).
nftables passé en « installé manuellement ».
0 mis à jour, 0 nouvellement installés, 0 à enlever et 9 non mis à jour.
doche@ubuntuserve:~$ sudo systemctl enable nftables
Created symlink /etc/systemd/system/sysinit.target.wants/nftables.service →


doche@ubuntuserve:~$
doche@ubuntuserve:~$ sudo nano /etc/nftables.conf
```

```
doche@ubuntuserve:~$ sudo nft -c -f /etc/nftables.conf
/etc/nftables.conf:34:2-5: Error: syntax error, unexpected string, expecting end of
 file or newline or semicolon
}sudo systemctl start nginx
 ^^^^

doche@ubuntuserve:~$ sudo systemctl restart nftables
doche@ubuntuserve:~$
```

5. Tests de sécurité

```
doche@ubuntuserve:~$ sudo apt update
Atteint :1 http://archive.ubuntu.com/ubuntu noble InRelease
Atteint :2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Atteint :3 http://security.ubuntu.com/ubuntu noble-security InRelease
Atteint :4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Atteint :5 https://repo.zabbix.com/zabbix-tools/debian-ubuntu noble InRelease
Atteint :6 https://repo.zabbix.com/zabbix/7.0/ubuntu noble InRelease
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
9 paquets peuvent être mis à jour. Exécutez « apt list --upgradable » pour les voir
.
doche@ubuntuserve:~$ sudo apt install nmap -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les paquets supplémentaires suivants seront installés :
  libblas3 liblinear4 libssh2-1t64 nmap-common
Paquets suggérés :
  liblinear-tools liblinear-dev ncat ndiff zenmap
Les NOUVEAUX paquets suivants seront installés :
  libblas3 liblinear4 libssh2-1t64 nmap nmap-common
0 mis à jour, 5 nouvellement installés, 0 à enlever et 9 non mis à jour.
Il est nécessaire de prendre 6 286 ko dans les archives.
Après cette opération, 27,4 Mo d'espace disque supplémentaires seront utilisés.
Réception de :1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libblas3
amd64 3.12.0-3build1.1 [238 kB]
Réception de :2 http://archive.ubuntu.com/ubuntu noble/universe amd64 liblinear4 am
d64 2.3.0+dfsg-5build1 [42,3 kB]
Réception de :3 http://archive.ubuntu.com/ubuntu noble/main amd64 libssh2-1t64 amd6
4 1.11.0-4.1build2 [120 kB]
Réception de :4 http://archive.ubuntu.com/ubuntu noble/universe amd64 nmap-common a
ll 7.94+git20230807.3be01efb1+dfsg-3build2 [4 192 kB]
Réception de :5 http://archive.ubuntu.com/ubuntu noble/universe amd64 nmap amd64 7.
94+git20230807.3be01efb1+dfsg-3build2 [1 694 kB]
6 286 ko réceptionnés en 4s (1 786 ko/s)
```

```
doche@ubuntuserve:~$ nmap localhost
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-22 01:36 UTC
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00012s latency).
Not shown: 994 closed tcp ports (conn-refused)
PORT     STATE SERVICE
22/tcp   open  ssh
25/tcp   open  smtp
80/tcp   open  http
111/tcp  open  rpcbind
3306/tcp open  mysql
8080/tcp open  http-proxy

Nmap done: 1 IP address (1 host up) scanned in 0.39 seconds
doche@ubuntuserve:~$
```

Durcissement d’un serveur OpenSSH

6. Analyse de la configuration actuelle

doche@ubuntuserve: ~

```
doche@ubuntuserve:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Sun 2026-03-22 00:51:16 UTC; 49min ago
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 840 (sshd)
      Tasks: 1 (limit: 4601)
     Memory: 4.0M (peak: 5.5M)
        CPU: 277ms
     CGroup: /system.slice/ssh.service
             └─840 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

mars 22 00:51:12 ubuntuserve systemd[1]: Starting ssh.service - OpenBSD Secure She>
mars 22 00:51:16 ubuntuserve sshd[840]: Server listening on 0.0.0.0 port 22.
mars 22 00:51:16 ubuntuserve sshd[840]: Server listening on :: port 22.
mars 22 00:51:16 ubuntuserve systemd[1]: Started ssh.service - OpenBSD Secure Shel>
mars 22 00:56:27 ubuntuserve sshd[1511]: Accepted password for doche from 172.20.1>
mars 22 00:56:27 ubuntuserve sshd[1511]: pam_unix(sshd:session): session opened fo>
lines 1-19/19 (END)
[4]+  Stopped                 systemctl status ssh


doche@ubuntuserve:~$ sudo cat /etc/ssh/sshd_config

# This is the sshd server system-wide configuration file.  See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bi
n:/sbin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented.  Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (the default), the socket
# configuration must be re-generated after changing Port, AddressFamily, or
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
#Port 22
#AddressFamily any
```

doche@ubuntuserve: ~

```
#   systemctl restart ssh.socket
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile     .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
#PasswordAuthentication yes
#PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
```

doche@ubuntuserve: ~

```
# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

# GSSAPI options
#GSSAPIAuthentication no
#GSSAPICleanupCredentials yes
#GSSAPIStrictAcceptorCheck yes
#GSSAPIKeyExchange no

# Set this to 'yes' to enable PAM authentication, account processing,
# and session processing. If this is enabled, PAM authentication will
# be allowed through the KbdInteractiveAuthentication and
# PasswordAuthentication.  Depending on your PAM configuration,
# PAM authentication via KbdInteractiveAuthentication may bypass
# the setting of "PermitRootLogin prohibit-password".
# If you just want the PAM account and session checks to run without
# PAM authentication, then enable this but set PasswordAuthentication
# and KbdInteractiveAuthentication to 'no'.
UsePAM yes

#AllowAgentForwarding yes
#AllowTcpForwarding yes
#GatewayPorts no
X11Forwarding yes
#X11DisplayOffset 10
#X11UseLocalhost yes
#PermitTTY yes
PrintMotd no
#PrintLastLog yes
#TCPKeepAlive yes
#PermitUserEnvironment no
#Compression delayed
#ClientAliveInterval 0
#ClientAliveCountMax 3
#UseDNS no
#PidFile /run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
#Banner none

# Allow client to pass locale environment variables
AcceptEnv LANG LC_*
```

```
doche@ubuntuserve: ~

# and session processing. If this is enabled, PAM authentication will
# be allowed through the KbdInteractiveAuthentication and
# PasswordAuthentication.  Depending on your PAM configuration,
# PAM authentication via KbdInteractiveAuthentication may bypass
# the setting of "PermitRootLogin prohibit-password".
# If you just want the PAM account and session checks to run without
# PAM authentication, then enable this but set PasswordAuthentication
# and KbdInteractiveAuthentication to 'no'.
UsePAM yes

#AllowAgentForwarding yes
#AllowTcpForwarding yes
#GatewayPorts no
X11Forwarding yes
#X11DisplayOffset 10
#X11UseLocalhost yes
#PermitTTY yes
PrintMotd no
#PrintLastLog yes
#TCPKeepAlive yes
#PermitUserEnvironment no
#Compression delayed
#ClientAliveInterval 0
#ClientAliveCountMax 3
#UseDNS no
#PidFile /run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
#Banner none

# Allow client to pass locale environment variables
AcceptEnv LANG LC_*

# override default of no subsystems
Subsystem       sftp    /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#       X11Forwarding no
#       AllowTcpForwarding no
#       PermitTTY no
#       ForceCommand cvs server
PermitRootLogin no
PasswordAuthentication yes

doche@ubuntuserve:~$
```

7. Désactiver la connexion root

```
  GNU nano 7.2                    /etc/ssh/sshd_config

# This is the sshd server system-wide configuration file.  See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/b>

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented.  Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (the default), the socket
# configuration must be re-generated after changing Port, AddressFamily, or
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

                              [ Read 134 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut     ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste   ^J Justify   ^/ Go To Line
```

```
doche@ubuntuserve:~$ sudo systemctl restart ssh
doche@ubuntuserve:~$
```

8. Authentification par clé SSH

```
doche@ubuntuserve:~$ ^C
doche@ubuntuserve:~$ chmod 700 ~/.ssh
doche@ubuntuserve:~$ chmod 600 ~/.ssh/authorized_keys
doche@ubuntuserve:~$
```

```
Windows PowerShell                    doche@ubuntuserve: ~

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka
.ms/PSWindows

PS C:\Users\dochl> ssh doche@172.20.10.2
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-106-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of dim. 22 mars 2026 02:18:17 UTC

  System load:  0.11               Processes:               130
  Usage of /:   33.5% of 24.44GB   Users logged in:         1
  Memory usage: 16%                IPv4 address for enp0s3: 172.20.10.2
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8
s
   just raised the bar for easy, resilient and secure K8s cluster deployment
.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

La maintenance de sécurité étendue pour Applications n'est pas activée.

9 mises à jour peuvent être appliquées immédiatement.
8 de ces mises à jour sont des mises à jour de sécurité.
Pour afficher ces mises à jour supplémentaires, exécuter : apt list --upgrad
able

1 mise à jour de sécurité supplémentaire peut être appliquée avec ESM Apps
En savoir plus sur l'activation du service ESM Apps at https://ubuntu.com/es
m


You have new mail.
Last login: Sun Mar 22 00:56:29 2026 from 172.20.10.4
doche@ubuntuserve:~$
```

9.Désactiver l’authentification par mot de passe

```
Windows PowerShell        doche@ubuntuserve: ~

  GNU nano 7.2                  /etc/ssh/sshd_config *

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin>

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented.  Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (the default), the socket
# configuration must be re-generated after changing Port, AddressFamily, or
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

```
doche@ubuntuserve:~$ sudo nano /etc/ssh/sshd_config
[sudo] password for doche:
doche@ubuntuserve:~$ sudo systemctl restart ssh
doche@ubuntuserve:~$
```

10. Changer le port SSH

```
doche@ubuntuserve:~$ sudo ufw allow 2222/tcp
ERROR: problem running
doche@ubuntuserve:~$ sudo nft flush ruleset
doche@ubuntuserve:~$ sudo ufw allow 2222/tcp
ERROR: problem running
doche@ubuntuserve:~$ sudo ufw disable
Firewall stopped and disabled on system startup
doche@ubuntuserve:~$ sudo systemctl restart ufw
doche@ubuntuserve:~$ sudo ufw allow 2222/tcp
Skipping adding existing rule
Skipping adding existing rule (v6)
doche@ubuntuserve:~$ sudo ufw status
Status: inactive
doche@ubuntuserve:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)?
y
Firewall is active and enabled on system startup
doche@ubuntuserve:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
22                         ALLOW IN    Anywhere
80                         ALLOW IN    Anywhere
443                        ALLOW IN    Anywhere
2222/tcp                   ALLOW IN    Anywhere
22/tcp (v6)                ALLOW IN    Anywhere (v6)
22 (v6)                    ALLOW IN    Anywhere (v6)
80 (v6)                    ALLOW IN    Anywhere (v6)
443 (v6)                   ALLOW IN    Anywhere (v6)
2222/tcp (v6)              ALLOW IN    Anywhere (v6)

doche@ubuntuserve:~$
```

```
doche@ubuntuserve:~$ sudo sshd -t
[sudo] password for doche:
doche@ubuntuserve:~$ grep -i "port" /etc/ssh/sshd_config
# configuration must be re-generated after changing Port, AddressFamily, or
Port 2222
#GatewayPorts no
doche@ubuntuserve:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: >
     Active: active (running) since Sun 2026-03-22 02:45:56 UTC; 3min 16s a>
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 4269 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUC>
   Main PID: 4272 (sshd)
      Tasks: 1 (limit: 4601)
     Memory: 3.0M (peak: 3.7M)
        CPU: 168ms
     CGroup: /system.slice/ssh.service
             └─4272 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startup>

mars 22 02:45:56 ubuntuserve systemd[1]: Starting ssh.service - OpenBSD Sec>
mars 22 02:45:56 ubuntuserve sshd[4272]: Server listening on 0.0.0.0 port 2>
mars 22 02:45:56 ubuntuserve sshd[4272]: Server listening on :: port 22.
mars 22 02:45:56 ubuntuserve systemd[1]: Started ssh.service - OpenBSD Secu>
mars 22 02:47:37 ubuntuserve sshd[4276]: Accepted publickey for doche from >
mars 22 02:47:37 ubuntuserve sshd[4276]: pam_unix(sshd:session): session op>
lines 1-20/20 (END)
[1]+  Stopped                 sudo systemctl status ssh
doche@ubuntuserve:~$ sudo systemctl stop ssh.socket
sudo systemctl disable ssh.socket
Removed "/etc/systemd/system/ssh.service.requires/ssh.socket".
Removed "/etc/systemd/system/sockets.target.wants/ssh.socket".
doche@ubuntuserve:~$ sudo systemctl stop ssh.socket
doche@ubuntuserve:~$ sudo systemctl disable ssh.socket
doche@ubuntuserve:~$ sudo systemctl restart ssh
doche@ubuntuserve:~$ ss -tulpn | grep 2222
tcp   LISTEN 0      128                                0.0.0.0:2222       0.0.
0.0:*
tcp   LISTEN 0      128                                   [::]:2222

doche@ubuntuserve:~$ sudo systemctl daemon-reload
doche@ubuntuserve:~$ sudo systemctl restart ssh.socket
```

Windows | Windows | doche@u | Windows

```
  GNU nano 7.2        /lib/systemd/system/ssh.socket *
[Unit]
Description=OpenBSD Secure Shell server socket
Before=sockets.target ssh.service
ConditionPathExists=!/etc/ssh/sshd_not_to_be_run

[Socket]
ListenStream=0.0.0.0:2222
ListenStream=[::]:2222
BindIPv6Only=ipv6-only
Accept=no
FreeBind=yes

[Install]
WantedBy=sockets.target
RequiredBy=ssh.service
```

```
     CGroup: /system.slice/ssh.service
             └─4272 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startup>

mars 22 02:45:56 ubuntuserve systemd[1]: Starting ssh.service - OpenBSD Sec>
mars 22 02:45:56 ubuntuserve sshd[4272]: Server listening on 0.0.0.0 port 2>
mars 22 02:45:56 ubuntuserve sshd[4272]: Server listening on :: port 22.
mars 22 02:45:56 ubuntuserve systemd[1]: Started ssh.service - OpenBSD Secu>
mars 22 02:47:37 ubuntuserve sshd[4276]: Accepted publickey for doche from >
mars 22 02:47:37 ubuntuserve sshd[4276]: pam_unix(sshd:session): session op>
lines 1-20/20 (END)
[1]+  Stopped                 sudo systemctl status ssh
doche@ubuntuserve:~$ sudo systemctl stop ssh.socket
sudo systemctl disable ssh.socket
Removed "/etc/systemd/system/ssh.service.requires/ssh.socket".
Removed "/etc/systemd/system/sockets.target.wants/ssh.socket".
doche@ubuntuserve:~$ sudo systemctl stop ssh.socket
doche@ubuntuserve:~$ sudo systemctl disable ssh.socket
doche@ubuntuserve:~$ sudo systemctl restart ssh
doche@ubuntuserve:~$ ss -tulpn | grep 2222
tcp   LISTEN 0      128                              0.0.0.0:2222       0.0.
0.0:*
tcp   LISTEN 0      128                                 [::]:2222

doche@ubuntuserve:~$ sudo systemctl daemon-reload
doche@ubuntuserve:~$ sudo systemctl restart ssh.socket
Job failed. See "journalctl -xe" for details.
doche@ubuntuserve:~$ sudo nano /etc/ssh/sshd_config









doche@ubuntuserve:~$ sudo systemctl daemon-reload
doche@ubuntuserve:~$ sudo systemctl stop ssh.service
doche@ubuntuserve:~$ sudo systemctl stop ssh.socket
doche@ubuntuserve:~$ sudo systemctl start ssh.socket
doche@ubuntuserve:~$ ss -tulpn | grep 2222
tcp   LISTEN 0      4096                             0.0.0.0:2222       0.0.
0.0:*
tcp   LISTEN 0      4096                                [::]:2222          [
::]:*
doche@ubuntuserve:~$
```

La connexion :

```
PS C:\Users\dochl> ssh -p 2222 doche@172.20.10.2
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-106-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of dim. 22 mars 2026 02:59:43 UTC

  System load:  0.17               Processes:               139
  Usage of /:   33.5% of 24.44GB   Users logged in:         1
  Memory usage: 16%                IPv4 address for enp0s3: 172.20.10.2
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8
s
   just raised the bar for easy, resilient and secure K8s cluster deployment
.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

La maintenance de sécurité étendue pour Applications n'est pas activée.

9 mises à jour peuvent être appliquées immédiatement.
8 de ces mises à jour sont des mises à jour de sécurité.
Pour afficher ces mises à jour supplémentaires, exécuter : apt list --upgrad
able

1 mise à jour de sécurité supplémentaire peut être appliquée avec ESM Apps
En savoir plus sur l'activation du service ESM Apps at https://ubuntu.com/es
m


You have new mail.
Last login: Sun Mar 22 02:47:39 2026 from 172.20.10.4
doche@ubuntuserve:~$
```

11.Protection contre brute force

Windows | Windows | doche@u | doche@u

```
doche@ubuntuserve:~$ sudo apt install fail2ban -y
[sudo] password for doche:
Sorry, try again.
[sudo] password for doche:
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les paquets supplémentaires suivants seront installés :
  python3-pyasyncore python3-pyinotify whois
Paquets suggérés :
  monit sqlite3 python-pyinotify-doc
Les NOUVEAUX paquets suivants seront installés :
  fail2ban python3-pyasyncore python3-pyinotify whois
0 mis à jour, 4 nouvellement installés, 0 à enlever et 9 non mis à jour.
Il est nécessaire de prendre 496 ko dans les archives.
Après cette opération, 2 572 ko d'espace disque supplémentaires seront utili
sés.
Réception de :1 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-py
asyncore all 1.0.2-2 [10,1 kB]
Réception de :2 http://archive.ubuntu.com/ubuntu noble-updates/universe amd6
4 fail2ban all 1.0.2-3ubuntu0.1 [409 kB]
Réception de :3 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-py
inotify all 0.9.6-2ubuntu1 [25,0 kB]
Réception de :4 http://archive.ubuntu.com/ubuntu noble/main amd64 whois amd6
4 5.5.22 [51,7 kB]
496 ko réceptionnés en 1s (359 ko/s)
Sélection du paquet python3-pyasyncore précédemment désélectionné.
(Lecture de la base de données... 135282 fichiers et répertoires déjà instal
lés.)
Préparation du dépaquetage de .../python3-pyasyncore_1.0.2-2_all.deb ...
Dépaquetage de python3-pyasyncore (1.0.2-2) ...
Sélection du paquet fail2ban précédemment désélectionné.
Préparation du dépaquetage de .../fail2ban_1.0.2-3ubuntu0.1_all.deb ...
Dépaquetage de fail2ban (1.0.2-3ubuntu0.1) ...
Sélection du paquet python3-pyinotify précédemment désélectionné.
Préparation du dépaquetage de .../python3-pyinotify_0.9.6-2ubuntu1_all.deb .
..
Dépaquetage de python3-pyinotify (0.9.6-2ubuntu1) ...
Sélection du paquet whois précédemment désélectionné.
Préparation du dépaquetage de .../whois_5.5.22_amd64.deb ...
Dépaquetage de whois (5.5.22) ...
Paramétrage de whois (5.5.22) ...
Paramétrage de python3-pyasyncore (1.0.2-2) ...
Paramétrage de fail2ban (1.0.2-3ubuntu0.1) ...
/usr/lib/python3/dist-packages/fail2ban/tests/fail2banregextestcase.py:224:
SyntaxWarning: invalid escape sequence '\s'
```

```
Windows   Windows   doche@u   doche@u

Warning: invalid escape sequence '\s'
  "`{ nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr-set-j-w-n
ft-ap\s+.*\s+\Khandle\s+(\d+)$'; } | while read -r hdl; do`",
/usr/lib/python3/dist-packages/fail2ban/tests/servertestcase.py:1424: Syntax
Warning: invalid escape sequence '\s'
  "`{ nft -a list chain inet f2b-table f2b-chain | grep -oP '@addr6-set-j-w-
nft-ap\s+.*\s+\Khandle\s+(\d+)$'; } | while read -r hdl; do`",
Created symlink /etc/systemd/system/multi-user.target.wants/fail2ban.service
 → /usr/lib/systemd/system/fail2ban.service.
Paramétrage de python3-pyinotify (0.9.6-2ubuntu1) ...
Traitement des actions différées (« triggers ») pour man-db (2.12.0-4build2)
 ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
doche@ubuntuserve:~$ sudo systemctl status fail2ban
● fail2ban.service - Fail2Ban Service
     Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; pre>
     Active: active (running) since Sun 2026-03-22 03:04:11 UTC; 27s ago
       Docs: man:fail2ban(1)
   Main PID: 4920 (fail2ban-server)
      Tasks: 5 (limit: 4601)
     Memory: 32.3M (peak: 32.5M)
        CPU: 1.981s
     CGroup: /system.slice/fail2ban.service
             └─4920 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

mars 22 03:04:11 ubuntuserve systemd[1]: Started fail2ban.service - Fail2Ba>
mars 22 03:04:12 ubuntuserve fail2ban-server[4920]: 2026-03-22 03:04:12,247>
mars 22 03:04:15 ubuntuserve fail2ban-server[4920]: Server ready
lines 1-14/14 (END)
[1]+  Stopped                 sudo systemctl status fail2ban
doche@ubuntuserve:~$ sudo fail2ban-client status
Status
|- Number of jail:      1
`- Jail list:   sshd
doche@ubuntuserve:~$
```

12.Tests de sécurité

Vérifier les ports ouverts :

```
doche@ubuntuserve:~$ ss -tuln
Netid  State   Recv-Q   Send-Q                          Local Address:Port
                   Peer Address:Port                  Process
udp    UNCONN  0        0                                   127.0.0.1:161
                       0.0.0.0:*
udp    UNCONN  0        0                                  127.0.0.54:53
                       0.0.0.0:*
udp    UNCONN  0        0                                127.0.0.53%lo:53
                       0.0.0.0:*
udp    UNCONN  0        0                           172.20.10.2%enp0s3:68
                       0.0.0.0:*
udp    UNCONN  0        0                                     0.0.0.0:111
                       0.0.0.0:*
udp    UNCONN  0        0                                       [::1]:161
                          [::]:*
udp    UNCONN  0        0           [fe80::a00:27ff:fe62:e94a]%enp0s3:546
                          [::]:*
udp    UNCONN  0        0                                        [::]:111
                          [::]:*
tcp    LISTEN  0        100                                   0.0.0.0:25
                       0.0.0.0:*
tcp    LISTEN  0        4096                                  0.0.0.0:111
                       0.0.0.0:*
tcp    LISTEN  0        511                                   0.0.0.0:80
                       0.0.0.0:*
tcp    LISTEN  0        4096                                  0.0.0.0:2222
                       0.0.0.0:*
tcp    LISTEN  0        151                                 127.0.0.1:3306
                       0.0.0.0:*
tcp    LISTEN  0        4096                               127.0.0.54:53
                       0.0.0.0:*
tcp    LISTEN  0        4096                             127.0.0.53%lo:53
                       0.0.0.0:*
tcp    LISTEN  0        70                                  127.0.0.1:33060
                       0.0.0.0:*
tcp    LISTEN  0        4096                                  0.0.0.0:10050
                       0.0.0.0:*
tcp    LISTEN  0        100                                      [::]:25
                          [::]:*
tcp    LISTEN  0        4096                                     [::]:111
                          [::]:*
tcp    LISTEN  0        511                                      [::]:80
                          [::]:*
tcp    LISTEN  0        4096                                     [::]:2222
                          [::]:*
tcp    LISTEN  0        4096                                     [::]:10050
```

```
udp    UNCONN  0        0                                        [::]:111
                          [::]:*
tcp    LISTEN  0        100                                   0.0.0.0:25
                       0.0.0.0:*
tcp    LISTEN  0        4096                                  0.0.0.0:111
                       0.0.0.0:*
tcp    LISTEN  0        511                                   0.0.0.0:80
                       0.0.0.0:*
tcp    LISTEN  0        4096                                  0.0.0.0:2222
                       0.0.0.0:*
tcp    LISTEN  0        151                                 127.0.0.1:3306
                       0.0.0.0:*
tcp    LISTEN  0        4096                               127.0.0.54:53
                       0.0.0.0:*
tcp    LISTEN  0        4096                           127.0.0.53%lo:53
                       0.0.0.0:*
tcp    LISTEN  0        70                                  127.0.0.1:33060
                       0.0.0.0:*
tcp    LISTEN  0        4096                                  0.0.0.0:10050
                       0.0.0.0:*
tcp    LISTEN  0        100                                      [::]:25
                          [::]:*
tcp    LISTEN  0        4096                                     [::]:111
                          [::]:*
tcp    LISTEN  0        511                                      [::]:80
                          [::]:*
tcp    LISTEN  0        4096                                     [::]:2222
                          [::]:*
tcp    LISTEN  0        4096                                     [::]:10050
                          [::]:*
tcp    LISTEN  0        511                                         *:8080
                              *:*
doche@ubuntuserve:~$ nmap 172.20.10.2
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-22 03:34 UTC
Nmap scan report for ubuntuserve (172.20.10.2)
Host is up (0.00011s latency).
Not shown: 995 closed tcp ports (conn-refused)
PORT     STATE SERVICE
25/tcp   open  smtp
80/tcp   open  http
111/tcp  open  rpcbind
2222/tcp open  EtherNetIP-1
8080/tcp open  http-proxy

Nmap done: 1 IP address (1 host up) scanned in 0.44 seconds
doche@ubuntuserve:~$
```

doche@ubuntuserve: ~

```
doche@ubuntuserve:~$ sudo ufw status
[sudo] password for doche:
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22                         ALLOW       Anywhere
80                         ALLOW       Anywhere
443                        ALLOW       Anywhere
2222/tcp                   ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
22 (v6)                    ALLOW       Anywhere (v6)
80 (v6)                    ALLOW       Anywhere (v6)
443 (v6)                   ALLOW       Anywhere (v6)
2222/tcp (v6)              ALLOW       Anywhere (v6)

doche@ubuntuserve:~$ sudo iptables -L
Chain INPUT (policy DROP)
target     prot opt source               destination
ufw-before-logging-input  all  --  anywhere             anywhere
ufw-before-input  all  --  anywhere             anywhere
ufw-after-input  all  --  anywhere             anywhere
ufw-after-logging-input  all  --  anywhere             anywhere
ufw-reject-input  all  --  anywhere             anywhere
ufw-track-input  all  --  anywhere             anywhere

Chain FORWARD (policy DROP)
target     prot opt source               destination
ufw-before-logging-forward  all  --  anywhere             anywhere
ufw-before-forward  all  --  anywhere             anywhere
ufw-after-forward  all  --  anywhere             anywhere
ufw-after-logging-forward  all  --  anywhere             anywhere
ufw-reject-forward  all  --  anywhere             anywhere
ufw-track-forward  all  --  anywhere             anywhere

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
ufw-before-logging-output  all  --  anywhere             anywhere
ufw-before-output  all  --  anywhere             anywhere
ufw-after-output  all  --  anywhere             anywhere
ufw-after-logging-output  all  --  anywhere             anywhere
ufw-reject-output  all  --  anywhere             anywhere
ufw-track-output  all  --  anywhere             anywhere

Chain ufw-after-forward (1 references)
```

```
doche@ubuntuserve:~$ sudo nft list ruleset
table inet filter {
        chain input {
                type filter hook input priority filter; policy accept;
        }

        chain forward {
                type filter hook forward priority filter; policy accept;
        }

        chain output {
                type filter hook output priority filter; policy accept;
        }
}
# Warning: table ip filter is managed by iptables-nft, do not touch!
table ip filter {
        chain ufw-before-logging-input {
        }

        chain ufw-before-logging-output {
        }

        chain ufw-before-logging-forward {
        }

        chain ufw-before-input {
                iifname "lo" counter packets 144 bytes 10305 accept
                ct state related,established counter packets 587 bytes 65799 accept
                ct state invalid counter packets 0 bytes 0 jump ufw-logging-deny
                ct state invalid counter packets 0 bytes 0 drop
                ip protocol icmp icmp type destination-unreachable counter packets 0 bytes 0
 accept
                ip protocol icmp icmp type time-exceeded counter packets 0 bytes 0 accept
                ip protocol icmp icmp type parameter-problem counter packets 0 bytes 0 accep
t
                ip protocol icmp icmp type echo-request counter packets 0 bytes 0 accept
                udp sport 67 udp dport 68 counter packets 3 bytes 984 accept
                counter packets 96 bytes 12447 jump ufw-not-local
                ip daddr 224.0.0.251 udp dport 5353 counter packets 0 bytes 0 accept
                ip daddr 239.255.255.250 udp dport 1900 counter packets 0 bytes 0 accept
                counter packets 96 bytes 12447 jump ufw-user-input
        }

        chain ufw-before-output {
                oifname "lo" counter packets 144 bytes 10305 accept
```

```
doche@ubuntuserve:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-03-26 20:38:47 UTC; 10min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 920 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 954 (sshd)
      Tasks: 1 (limit: 4601)
     Memory: 4.0M (peak: 5.0M)
        CPU: 54ms
     CGroup: /system.slice/ssh.service
             └─954 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

mars 26 20:38:46 ubuntuserve systemd[1]: Starting ssh.service - OpenBSD Secure Shell server>
mars 26 20:38:46 ubuntuserve sshd[954]: Server listening on 0.0.0.0 port 22.
mars 26 20:38:46 ubuntuserve sshd[954]: Server listening on :: port 22.
```

```
doche@ubuntuserve:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-03-26 20:38:47 UTC; 11min ago
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 920 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 954 (sshd)
      Tasks: 1 (limit: 4601)
     Memory: 4.0M (peak: 5.0M)
        CPU: 54ms
     CGroup: /system.slice/ssh.service
             └─954 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

1. Différence entre firewall stateful et stateless ?

La différence entre firewall stateful et stateless ?

Le firewall Stateless

Fonctionnement examine chaque paquet de données de manière indépendante, sans tenir compte des paquets précédents ou suivants.

Avantages :simple, rapide, peu gourmand en ressources.

inconvénients :moins sécurisé, car il ne peut pas détecter les attaques basées sur le contexte (comme le spoofing ou les attaques par déni de service).

Tansdis que Firewall Stateful

Fonctionnement suit l'état des connexions réseau (comme TCP, UDP) et prend des décisions en fonction du contexte (par exemple, il se souvient qu'une connexion a été initiée de l'intérieur vers l'extérieur).

Avantages :plus sécurisé, car il peut détecter des comportements anormaux et bloquer des attaques plus complexes.

Inconvénients plus complexe, nécessite plus de ressources (mémoire, CPU).

Le firewall stateful est plus intelligent et sécurisé, tandis que le stateless est plus basique et rapide.

2. Avantages de nftables par rapport à iptables ?

R)Les avantages de nftables par rapport à iptables sont : syntaxe unifiée une seule commande (nft) pour ipv4, ipv6, arp, etc.

Les performances moins de duplication de règles, traitement plus rapide.

La simplicité tables, chaînes et règles sont gérées de manière cohérente.
L'extensibilité support natif des ensembles (sets) et des cartes (maps).
La compatibilité peut émuler iptables via un layer de compatibilité.

3. Pourquoi toujours autoriser SSH avant d'activer un firewall ?

R) le SSH sont toujours autoriser avant d'activer un firewall ?
Si tu actives le firewall avant d'autoriser SSH, tu risques de te bloquer toi-même l'accès à distance. Toujours vérifier que la règle permettant SSH (port 22 par défaut) est en place avant d'appliquer les autres règles.

4. Pourquoi désactiver la connexion root ?
R)La connexion root est désactiver parce que la sécurité limite les attaques par force brute sur le compte root.
La traçabilité Oblige à utiliser un utilisateur normal puis sudo, ce qui permet de mieux auditer les actions.
Les Bonnes pratiques Réduit les risques d'erreur ou de compromission du compte le plus puissant.

5. Avantages de changer le port SSH ?
R)Les Avantages de changer le port SSH sont
La réduction du bruit : moins de tentatives de connexion automatiques (bots scannant le port 22).
La sécurité par l'obscurité pas une protection absolue, mais complique la vie des attaquants occasionnels.
La journalisation plus claire moins de logs inutiles.

6. Quel est le rôle de Fail2Ban ?

R)Le Rôle de Fail2Ban

Fail2Ban scanne les logs ssh, apache, etc. et bloque automatiquement les IP après un nombre défini de tentatives infructueuses. Utile contre les attaques par force brute.

À travers ce TD, j'ai pu explorer de manière concrète les différentes facettes de la sécurité réseau sous Linux. J'ai

commencé par analyser les ports ouverts afin de comprendre quels services étaient exposés. Ensuite, j'ai configuré un firewall simple et écrit des règles de filtrage pour contrôler le trafic. L'étude des chaînes de filtrage m'a permis de saisir la logique interne du pare-feu et son fonctionnement.

Ce TD m'a donné une vision complète et pratique des mécanismes de sécurisation réseau et des accès. J'ai acquis des compétences utiles pour réduire la surface d'attaque, protéger les services critiques et mettre en place une politique de sécurité cohérente et efficace.